In [25]:
```
import torch
from torch.autograd import Variable
import torch.nn as nn
import torch.nn.functional as F

import torch.optim as optim
import matplotlib.pyplot as plt

class Net(nn.Module):
    def __init__(self):
        super(Net, self).__init__()
        self.fc1 = nn.Linear(2,1,bias=False)
    def forward(self, x):
        x = self.fc1(x)
        return x

net = Net()
print(net)


net.fc1.weight = torch.nn.Parameter(torch.tensor([[1., -1.]], requires_grad=True

print(list(net.parameters()))

#input = torch.randn(1,2)
#out = net(input)


criterion = nn.MSELoss()
#def criterion(out, label):
#    return ((label - out)**2).mean()


optimizer = optim.SGD(net.parameters(), lr=0.01, momentum=0.5)
#optimizer = optim.Adam(net.parameters(), lr=0.005)


data = torch.tensor([[1.,3.], [2.,6.], [3.,9.]], dtype=torch.float)
target = torch.tensor([[1.],[5.],[13.]], dtype=torch.float)

hist = []

############### SGD-based update ###############

for epoch in range(300):
    epoch_loss = 0.0

    for i in range(len(data)):
        optimizer.zero_grad()
        output = net(data[i])
        loss = criterion(output, target[i])
        loss.backward()
        optimizer.step()

        epoch_loss += loss.item()

    # Average loss across all data points for this epoch
    avg_loss = epoch_loss / len(data)
```

```
    hist.append(loss.item())

    print(f"Epoch {epoch} - average loss: {avg_loss:.6f}")

################################################

# ############## Batch GD based update ##############

# for epoch in range(300):
#     optimizer.zero_grad()
#     outputs = net(data)
#     loss = criterion(outputs, target)
#     loss.backward()
#     hist.append(loss.detach())
#     optimizer.step()
#     print("Epoch {} - loss: {}".format(epoch, loss))
# ################################################

### Test the trained network ###
for i, current_data in enumerate(data):
    out = net(current_data)
    print("when x = {}, y = {}".format(current_data, out))

plt.plot(hist, label = "training curve")
```

```
Net(
  (fc1): Linear(in_features=2, out_features=1, bias=False)
)
[Parameter containing:
tensor([[ 1., -1.]], requires_grad=True)]
Epoch 0 - average loss: 39.334532
Epoch 1 - average loss: 80.625629
Epoch 2 - average loss: 37.829956
Epoch 3 - average loss: 75.121512
Epoch 4 - average loss: 37.674733
Epoch 5 - average loss: 74.591525
Epoch 6 - average loss: 37.395082
Epoch 7 - average loss: 74.140797
Epoch 8 - average loss: 37.116333
Epoch 9 - average loss: 73.694979
Epoch 10 - average loss: 36.840476
Epoch 11 - average loss: 73.252864
Epoch 12 - average loss: 36.567522
Epoch 13 - average loss: 72.814407
Epoch 14 - average loss: 36.297446
Epoch 15 - average loss: 72.379595
Epoch 16 - average loss: 36.030199
Epoch 17 - average loss: 71.948314
Epoch 18 - average loss: 35.765771
Epoch 19 - average loss: 71.520611
Epoch 20 - average loss: 35.504124
Epoch 21 - average loss: 71.096446
Epoch 22 - average loss: 35.245239
Epoch 23 - average loss: 70.675761
Epoch 24 - average loss: 34.989076
Epoch 25 - average loss: 70.258529
Epoch 26 - average loss: 34.735609
Epoch 27 - average loss: 69.844735
Epoch 28 - average loss: 34.484837
Epoch 29 - average loss: 69.434354
Epoch 30 - average loss: 34.236708
Epoch 31 - average loss: 69.027334
Epoch 32 - average loss: 33.991201
Epoch 33 - average loss: 68.623627
Epoch 34 - average loss: 33.748307
Epoch 35 - average loss: 68.223287
Epoch 36 - average loss: 33.507994
Epoch 37 - average loss: 67.826226
Epoch 38 - average loss: 33.270226
Epoch 39 - average loss: 67.432434
Epoch 40 - average loss: 33.034979
Epoch 41 - average loss: 67.041878
Epoch 42 - average loss: 32.802236
Epoch 43 - average loss: 66.654501
Epoch 44 - average loss: 32.571973
Epoch 45 - average loss: 66.270310
Epoch 46 - average loss: 32.344158
Epoch 47 - average loss: 65.889284
Epoch 48 - average loss: 32.118780
Epoch 49 - average loss: 65.511377
Epoch 50 - average loss: 31.895798
Epoch 51 - average loss: 65.136550
Epoch 52 - average loss: 31.675208
Epoch 53 - average loss: 64.764818
Epoch 54 - average loss: 31.456959
```

```
Epoch 55 - average loss: 64.396093
Epoch 56 - average loss: 31.241049
Epoch 57 - average loss: 64.030391
Epoch 58 - average loss: 31.027448
Epoch 59 - average loss: 63.667667
Epoch 60 - average loss: 30.816141
Epoch 61 - average loss: 63.307920
Epoch 62 - average loss: 30.607109
Epoch 63 - average loss: 62.951150
Epoch 64 - average loss: 30.400318
Epoch 65 - average loss: 62.597258
Epoch 66 - average loss: 30.195754
Epoch 67 - average loss: 62.246260
Epoch 68 - average loss: 29.993370
Epoch 69 - average loss: 61.898108
Epoch 70 - average loss: 29.793171
Epoch 71 - average loss: 61.552782
Epoch 72 - average loss: 29.595129
Epoch 73 - average loss: 61.210298
Epoch 74 - average loss: 29.399229
Epoch 75 - average loss: 60.870614
Epoch 76 - average loss: 29.205450
Epoch 77 - average loss: 60.533667
Epoch 78 - average loss: 29.013749
Epoch 79 - average loss: 60.199490
Epoch 80 - average loss: 28.824132
Epoch 81 - average loss: 59.868010
Epoch 82 - average loss: 28.636563
Epoch 83 - average loss: 59.539219
Epoch 84 - average loss: 28.451031
Epoch 85 - average loss: 59.213117
Epoch 86 - average loss: 28.267512
Epoch 87 - average loss: 58.889658
Epoch 88 - average loss: 28.085991
Epoch 89 - average loss: 58.568835
Epoch 90 - average loss: 27.906438
Epoch 91 - average loss: 58.250619
Epoch 92 - average loss: 27.728834
Epoch 93 - average loss: 57.934971
Epoch 94 - average loss: 27.553174
Epoch 95 - average loss: 57.621890
Epoch 96 - average loss: 27.379428
Epoch 97 - average loss: 57.311344
Epoch 98 - average loss: 27.207589
Epoch 99 - average loss: 57.003342
Epoch 100 - average loss: 27.037636
Epoch 101 - average loss: 56.697848
Epoch 102 - average loss: 26.869532
Epoch 103 - average loss: 56.394802
Epoch 104 - average loss: 26.703272
Epoch 105 - average loss: 56.094210
Epoch 106 - average loss: 26.538822
Epoch 107 - average loss: 55.796036
Epoch 108 - average loss: 26.376185
Epoch 109 - average loss: 55.500264
Epoch 110 - average loss: 26.215350
Epoch 111 - average loss: 55.206948
Epoch 112 - average loss: 26.056284
Epoch 113 - average loss: 54.915959
Epoch 114 - average loss: 25.898954
```

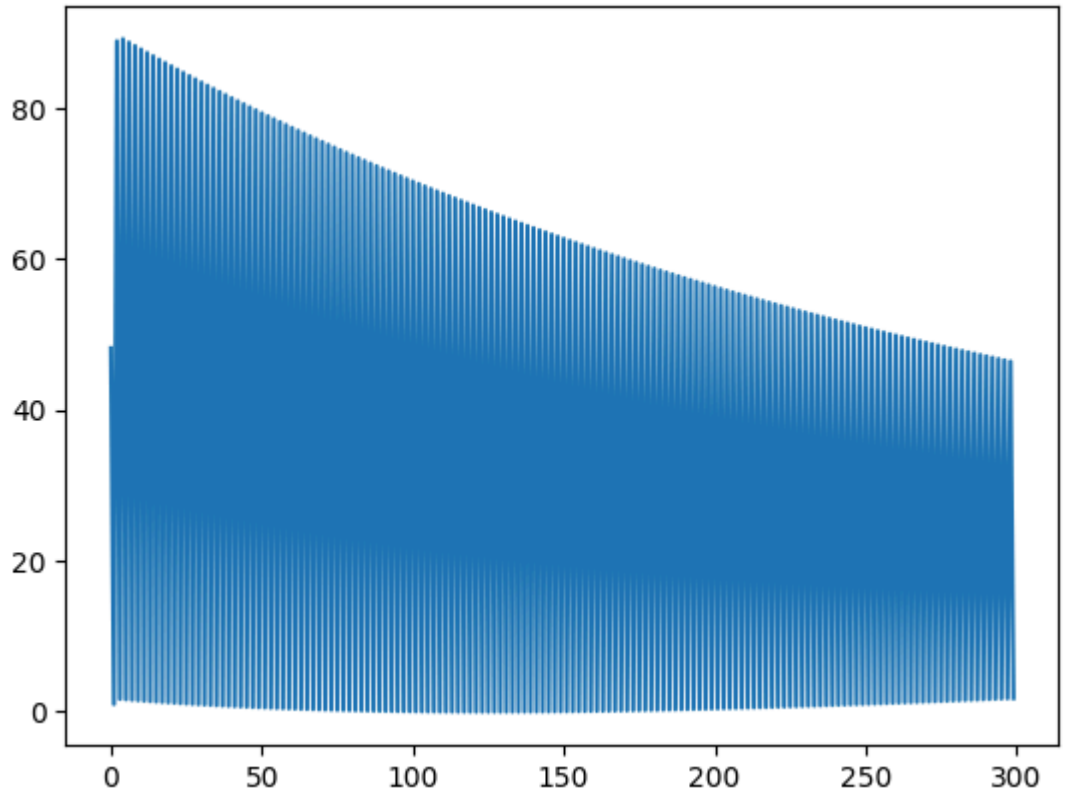
```
Epoch 115 - average loss: 54.627327
Epoch 116 - average loss: 25.743362
Epoch 117 - average loss: 54.341009
Epoch 118 - average loss: 25.589499
Epoch 119 - average loss: 54.057037
Epoch 120 - average loss: 25.437325
Epoch 121 - average loss: 53.775307
Epoch 122 - average loss: 25.286839
Epoch 123 - average loss: 53.495878
Epoch 124 - average loss: 25.138016
Epoch 125 - average loss: 53.218705
Epoch 126 - average loss: 24.990852
Epoch 127 - average loss: 52.943746
Epoch 128 - average loss: 24.845318
Epoch 129 - average loss: 52.671014
Epoch 130 - average loss: 24.701397
Epoch 131 - average loss: 52.400456
Epoch 132 - average loss: 24.559086
Epoch 133 - average loss: 52.132102
Epoch 134 - average loss: 24.418365
Epoch 135 - average loss: 51.865894
Epoch 136 - average loss: 24.279209
Epoch 137 - average loss: 51.601829
Epoch 138 - average loss: 24.141609
Epoch 139 - average loss: 51.339899
Epoch 140 - average loss: 24.005555
Epoch 141 - average loss: 51.080059
Epoch 142 - average loss: 23.871018
Epoch 143 - average loss: 50.822308
Epoch 144 - average loss: 23.737989
Epoch 145 - average loss: 50.566625
Epoch 146 - average loss: 23.606452
Epoch 147 - average loss: 50.312974
Epoch 148 - average loss: 23.476390
Epoch 149 - average loss: 50.061351
Epoch 150 - average loss: 23.347802
Epoch 151 - average loss: 49.811772
Epoch 152 - average loss: 23.220659
Epoch 153 - average loss: 49.564167
Epoch 154 - average loss: 23.094952
Epoch 155 - average loss: 49.318559
Epoch 156 - average loss: 22.970670
Epoch 157 - average loss: 49.074920
Epoch 158 - average loss: 22.847788
Epoch 159 - average loss: 48.833225
Epoch 160 - average loss: 22.726296
Epoch 161 - average loss: 48.593449
Epoch 162 - average loss: 22.606182
Epoch 163 - average loss: 48.355597
Epoch 164 - average loss: 22.487431
Epoch 165 - average loss: 48.119627
Epoch 166 - average loss: 22.370035
Epoch 167 - average loss: 47.885541
Epoch 168 - average loss: 22.253982
Epoch 169 - average loss: 47.653346
Epoch 170 - average loss: 22.139246
Epoch 171 - average loss: 47.422983
Epoch 172 - average loss: 22.025821
Epoch 173 - average loss: 47.194466
Epoch 174 - average loss: 21.913691
```

```
Epoch 175 - average loss: 46.967754
Epoch 176 - average loss: 21.802855
Epoch 177 - average loss: 46.742870
Epoch 178 - average loss: 21.693286
Epoch 179 - average loss: 46.519762
Epoch 180 - average loss: 21.584973
Epoch 181 - average loss: 46.298429
Epoch 182 - average loss: 21.477907
Epoch 183 - average loss: 46.078832
Epoch 184 - average loss: 21.372067
Epoch 185 - average loss: 45.860989
Epoch 186 - average loss: 21.267454
Epoch 187 - average loss: 45.644884
Epoch 188 - average loss: 21.164046
Epoch 189 - average loss: 45.430490
Epoch 190 - average loss: 21.061848
Epoch 191 - average loss: 45.217805
Epoch 192 - average loss: 20.960827
Epoch 193 - average loss: 45.006809
Epoch 194 - average loss: 20.860980
Epoch 195 - average loss: 44.797483
Epoch 196 - average loss: 20.762297
Epoch 197 - average loss: 44.589817
Epoch 198 - average loss: 20.664761
Epoch 199 - average loss: 44.383792
Epoch 200 - average loss: 20.568358
Epoch 201 - average loss: 44.179384
Epoch 202 - average loss: 20.473092
Epoch 203 - average loss: 43.976608
Epoch 204 - average loss: 20.378932
Epoch 205 - average loss: 43.775449
Epoch 206 - average loss: 20.285874
Epoch 207 - average loss: 43.575842
Epoch 208 - average loss: 20.193910
Epoch 209 - average loss: 43.377831
Epoch 210 - average loss: 20.103030
Epoch 211 - average loss: 43.181381
Epoch 212 - average loss: 20.013218
Epoch 213 - average loss: 42.986481
Epoch 214 - average loss: 19.924468
Epoch 215 - average loss: 42.793105
Epoch 216 - average loss: 19.836757
Epoch 217 - average loss: 42.601256
Epoch 218 - average loss: 19.750090
Epoch 219 - average loss: 42.410924
Epoch 220 - average loss: 19.664450
Epoch 221 - average loss: 42.222079
Epoch 222 - average loss: 19.579827
Epoch 223 - average loss: 42.034744
Epoch 224 - average loss: 19.496207
Epoch 225 - average loss: 41.848846
Epoch 226 - average loss: 19.413591
Epoch 227 - average loss: 41.664447
Epoch 228 - average loss: 19.331955
Epoch 229 - average loss: 41.481476
Epoch 230 - average loss: 19.251295
Epoch 231 - average loss: 41.299936
Epoch 232 - average loss: 19.171603
Epoch 233 - average loss: 41.119833
Epoch 234 - average loss: 19.092871
```

```
Epoch 235 - average loss: 40.941138
Epoch 236 - average loss: 19.015087
Epoch 237 - average loss: 40.763858
Epoch 238 - average loss: 18.938237
Epoch 239 - average loss: 40.587955
Epoch 240 - average loss: 18.862306
Epoch 241 - average loss: 40.413407
Epoch 242 - average loss: 18.787303
Epoch 243 - average loss: 40.240273
Epoch 244 - average loss: 18.713204
Epoch 245 - average loss: 40.068455
Epoch 246 - average loss: 18.640002
Epoch 247 - average loss: 39.897980
Epoch 248 - average loss: 18.567691
Epoch 249 - average loss: 39.728843
Epoch 250 - average loss: 18.496266
Epoch 251 - average loss: 39.561031
Epoch 252 - average loss: 18.425712
Epoch 253 - average loss: 39.394542
Epoch 254 - average loss: 18.356017
Epoch 255 - average loss: 39.229325
Epoch 256 - average loss: 18.287177
Epoch 257 - average loss: 39.065420
Epoch 258 - average loss: 18.219181
Epoch 259 - average loss: 38.902777
Epoch 260 - average loss: 18.152022
Epoch 261 - average loss: 38.741392
Epoch 262 - average loss: 18.085696
Epoch 263 - average loss: 38.581285
Epoch 264 - average loss: 18.020186
Epoch 265 - average loss: 38.422424
Epoch 266 - average loss: 17.955477
Epoch 267 - average loss: 38.264767
Epoch 268 - average loss: 17.891583
Epoch 269 - average loss: 38.108359
Epoch 270 - average loss: 17.828477
Epoch 271 - average loss: 37.953154
Epoch 272 - average loss: 17.766161
Epoch 273 - average loss: 37.799174
Epoch 274 - average loss: 17.704621
Epoch 275 - average loss: 37.646370
Epoch 276 - average loss: 17.643849
Epoch 277 - average loss: 37.494748
Epoch 278 - average loss: 17.583841
Epoch 279 - average loss: 37.344303
Epoch 280 - average loss: 17.524585
Epoch 281 - average loss: 37.195034
Epoch 282 - average loss: 17.466076
Epoch 283 - average loss: 37.046916
Epoch 284 - average loss: 17.408308
Epoch 285 - average loss: 36.899950
Epoch 286 - average loss: 17.351261
Epoch 287 - average loss: 36.754091
Epoch 288 - average loss: 17.294941
Epoch 289 - average loss: 36.609367
Epoch 290 - average loss: 17.239341
Epoch 291 - average loss: 36.465764
Epoch 292 - average loss: 17.184452
Epoch 293 - average loss: 36.323295
Epoch 294 - average loss: 17.130258
```

```
Epoch 295 - average loss: 36.181914
Epoch 296 - average loss: 17.076756
Epoch 297 - average loss: 36.041605
Epoch 298 - average loss: 17.023946
Epoch 299 - average loss: 35.902385
when x = tensor([1., 3.]), y = tensor([3.1382], grad_fn=<SqueezeBackward4>)
when x = tensor([2., 6.]), y = tensor([6.2764], grad_fn=<SqueezeBackward4>)
when x = tensor([3., 9.]), y = tensor([9.4147], grad_fn=<SqueezeBackward4>)
```

Out[25]: [<matplotlib.lines.Line2D at 0x236f978bf70>]



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